

The Effects of Cyber-Security on Healthcare Industry

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Abstract— Cyber-attacks today are targeting healthcare industries as the most preferred attacked industry. This paper introduced a comparative study of the different applications of cyber-security, and the changes in levels of risk for different industries. The study will focus on the use of cyber-security for healthcare industry, and the different techniques that are used to protect Internet of Things IoT-based healthcare industry. The study also reviews different types of security threats in healthcare industry. In this paper we are going to focus on the use of AES (Advanced Encryption Standard) for prtecting healthcare industry from cyber-attacks, and the application in Electronic Health Record (EHR).

Keywords—Cybersecurity, healthcare, IoT

I. INTRODUCTION

Nowadays Cyber-security became a trend in the field of research. It has different applications in many fields' like government services, military, education, financial services, healthcare applications, manufacturing, and transportation services. Previously, financial services were the highest at-risk industry, but in 2015, the healthcare industry became the top in cyber-attacks. As recorded in [1] that 100 million health records were breached in 2015 in a study that has been carried out for over 8,000 clients in 100 different countries. As shown in Fig. 1. that healthcare became the most attacked industry in 2015 while just in 2014 the financial services were the most attacked industry.



Fig. 1. Healthcare moved into the top spot the ranking as the most attacked industry in 2015 [1]

Most of Cyber-Security applications are Internet of Things based. Where IoT type of application faces the problem in terms of data exchange process where many challenges are in this field. Internet of Things (IoT) interconnects the cyberspace with physical world. In IoT healthcare applications the patient data are collected from different sources and gathered together in form of Electronic Health Record (EHR) to be used either by transferring them through the internet or by storing them on cloud. Electronic Health Record (EHR) is a systematic collection of electronic health information which defines patients. Also it is a digital format of records that can be shared between different healthcare settings [2]. It consists of different data such as demographics, medical history, medication and laboratory test results. Also it has some patients billing information [3]. Healthcare applications are very critical and needs to be strongly secured. Also healthcare industries are using IoT type of application and they are dynamic in their nature. So these types of applications are not using traditional security mechanism. However, not all mechanisms are totally newly produced, some traditional mechanisms are used with some modifications. Thus the healthcare industries have to be strongly secured and protected from any type of attacks or cyber-criminals. This paper aims to review the most common threats that face the healthcare industries and their countermeasures. Moreover this paper will discuss the use of AES algorithm in protecting Electronic Health Record from cyber-attacks. Advanced Encryption Standard algorithm has been considered as the fastest and most secure algorithm, as a result AES is chosen to be tested and implemented in order to be improved in terms of its design complexity by optimizing the hardware design. In this research we are using basic Rajdinel AES VHDL code for improvement and optimization in a hardware level [4]. The main focus of this paper is to improve and reduce the complexity of the substitution Byte component (S-box), where S-box is the most expansive component design as it is using a look up table operation instead of function operation which requires more time and

more design area since it is the only non-linear function. This paper is organized as follows. First section introduces the topic with some background, history and the aim of this paper. Then, second section highlights some literature reviews that focus on recent research for ensuring security and privacy of healthcare industries. Section three shows some of the common threats attacking healthcare industries. Section four shows some solutions as countermeasures for these threats to ensure security and privacy of the data in term of encrypting data. Section five shows Byte Substitution component simulation and utilization of AES algorithm. Finally, section five concludes this paper.

II. LITERATURE REVIEW

Due to the fast development of technology, many researches are introduced in the field to highlight the challenges and the issues. Also to propose solutions and algorithms to solve these issues and to face challenges. Since technology became in every field in all aspects of life. Many researches are proposed for ensuring the security and privacy of data in the field of IoT applications and especially in the field of healthcare applications.

Authors in [5] introduced scheme for end-to-end security that mobility enabled health care Internet of Things (IoT). As shown in Fig. 2. the proposed schema consists of architecture for authenticating and authorizing end users. Then, a secure end-to-end communication. Finally, a smart gateway that acts as a layer between sensor and cloud service. They simulate their schema and produced hardware and software for their proposed schema.

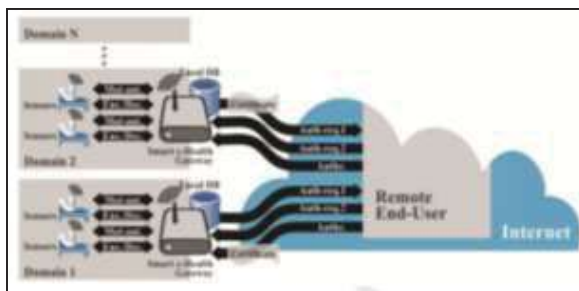


Fig. 2. The proposed SEA Architecture overview using distributed smart e-health [5]

This research results in reducing the overhead needed for the communication. Also they found that this schema is 97% faster than other schemas they compared it with.

In [6] authors developed a system used to ensure the security of modern IoT-based healthcare system that used body sensor network. This research highlights the security issues that faced the systems using body sensor network and they introduced a

system to meet these security requirements. They had a result where they lowered the execution time needed by 42% than other systems.

As introduced in [7], authors proposed an authentication schema for securing healthcare systems. It is basically two security protocols that have been proposed. The authentication schema for protecting IoT-based healthcare system and the coexistences proof schema used for multi tagged objects. Their result is that their schema ensured a robust secure communication. They implemented their schema to ensure the result.

A research proposed in [8], researchers developed a framework for centralized data that stored the data collected from several sensing units. This research aims to ensure the security, privacy and confidentiality of the system. They used two cryptographic methods. The attribute based and the functional encryption methods, where they were combined together. Fig. 3. shows the architecture of the proposed framework.

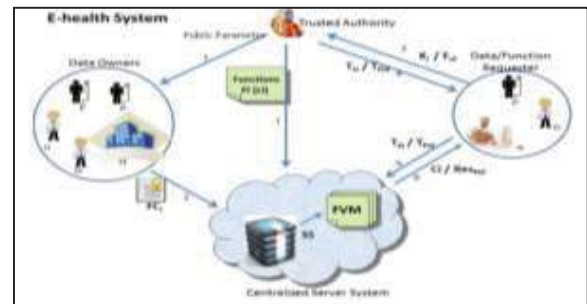


Fig. 3. Architecture of proposed framework [8]

Paper introduced in [9] proposed a cloud-based framework for secure healthcare application using Wireless Body Area Network (WBAN). They used to ensure the security of the inter-sensor communication by combining multi-biometric key generation schema. Also they used to ensure the privacy of the patient data by securing the EHR that centrally stored in health industry cloud. They experimentally tested their system and results showed that their system produced a secure framework that is cloud-based and ensured security and privacy of communication process and patient data.

Authors in [10] introduced a lightweight attribute-based Encryption (ABE) scheme to improve the ABE scheme that is popularly used in securing stored data, communication between devices and in sharing data in the IoT framework. Traditional (ABE) used to be considered an expansive schema in IoT framework. In their research they used elliptic curve cryptography (ECC) for data security and privacy. They tested their schema by some matrices to measure the communication

and computational overhead. Although the proposed schema has some limitations that were discussed, Results show that this schema was better in terms of cost and execution time than the existing schema.

III. SECURITY THREAT IN HEALTHCARE INDUSTRY

Healthcare applications are very critical applications and medical data are very critical and complex to be secure than other type of data and applications because it needs to be highly secured [11]. There are many types of threats that may face the healthcare applications which differ in their causes and differ in their solutions. Some of these security threats are highlighted in this paper. These security threats are eavesdropping, impersonation, message modification, and Man-in-the-middle.

A. Eavesdropping Attack

Using Alice, Bob and Attacker for the scenarios as actors. Eavesdropping attack evolves sniffing all the data sent from sender to receiver. So in the case of healthcare applications, sniffing these medical data will involve the privacy of the communication. Assuming Alice sends an unencrypted data to Bob, attacker may eavesdrop the data by sniffing it using any sniffing tools.

B. Impersonation

The impersonation is done by taking the identity of an authorized user and transmitting data with the IP address of this user without doing any authentication process.

C. Message Modification

The issue of message modification is evolved as follows, Alice sends some data, the attacker captures the transmitted data and modifies its content and resends it to receiver Bob using the IP address of Alice.

D. Man-In-The-Middle

Man-in-the-middle attack is evolved when the attacker establishes a connection between a legitimate sender and legitimate receiver independently in order to modify the messages between them without the knowledge of the real sender and receiver [12].

In order to prevent and secure healthcare applications from being attacked, different security mechanisms have to be used to prevent different types of security attacks. For example having good authentication mechanism and having a strong model for encryption and decryption of the medical data will secure the data from being attacked.

IV. SECURITY COUNTERMEASURES

There are several security countermeasures that are used to solve security threats. Different solutions are used to solve different types of threats. Some of the solutions are Encryption, Authentication and Authorization to protect the

data stored from different type of threats. This paper focuses on Encryption algorithms for protecting data from different type of security threats but not all of them where in some cases encryption and authentication method is needed for protecting data especially in the IoT framework as it is dynamic framework.

However, traditional type of security countermeasures cannot be applied directly to IoT type of application, they are used as a basis for newly produced methods. In this paper, some of the most common symmetric Encryption algorithms will be discussed. These algorithms are DES, 3DES, Blue fish and AES.

A. DES

Data Encryption Standard (DES) algorithm was produced in 1977 by IBM. Basically, this algorithm is used to encrypt a stream fixed-length of bits of plaintext. Then this plaintext is transferred to cipher-text of the same size. The size of the block is 64 bits where 56-bits for the algorithm as key length and the rest 8 bits are used for check party. This algorithm is considered as a very slow encryption algorithm [13].

B. Bluefish

This algorithm was produced in 1993. It uses block size of 64 bits and key size between 32 to 448 bits. It was introduced to alternatively replace DES algorithm. It uses a variable key size to be faster and more secure. It is a free license algorithm and it is freely available for all users. It is considered to be faster and more secure than DES and 3DES [14].

C. 3DES

In 1998 Triple Data Encryption Standard introduced as an improvement of DES. This algorithm applied the DES three times. It also uses the same number of block size 64 bits where 56 bits for key size. This algorithm is considered to be faster than DES but it is also considered to be a slow algorithm since it uses to apply DES three times. It is more secure than DES [15].

D. AES

Advanced Encryption Standard (AES) algorithm produced in 2001. It is organized by National Institute of Standards and Technology (NIST). It uses a block size of 128bits while it uses key size of 128,192 or 256 bits. The number of rounds based on the key size. As an example for 128 bits 10 rounds is needed, for 192 bits 12 rounds are used and for 256 bits it uses 14 rounds. This algorithm considered as a very fast algorithm with high security [16].

V. AES BYTE SUBSTITUTION SIMULATION AND UTILIZATION

The byte substitution unit in Rajdinel algorithm is considered to be the most complex component due to the design of S-box, the flow diagram of the Encryption process of Rajdinel algorithm is shown in Fig. 4,. it consists of a number of transformations and their sequence for encryption.

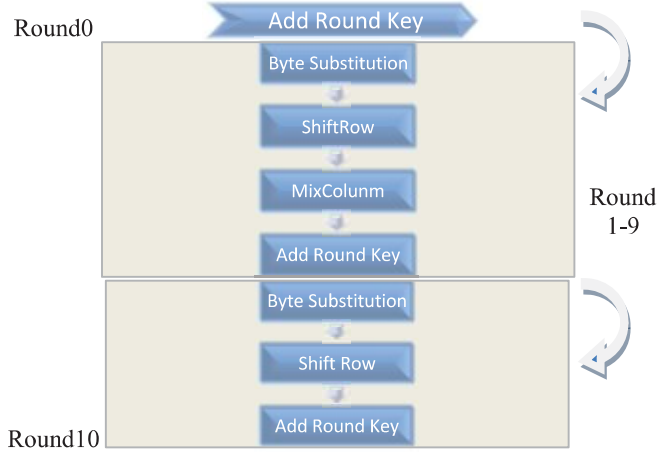


Fig. 4. Flow diagram of Rajdinel Encryption algorithm

This diagram shows the Rajdinel Encryption algorithm for 128-bits AES. It requires 10 rounds. First and last rounds are different while the middle nine rounds are the same. First step uses the Add Round Key function only. Round 1 to round 9 use different stages of transformation sub Byte, ShiftRow, MixColumn and Add Round Key. Last Round has the following functions Sub Byte, ShiftRow and finally Add Round Key.

Add Round Key (): this function uses XORing operation between the generating key and the output from the Mix column function.

Sub-Byte(): this is the only non-linear function in the algorithm. It is done through two steps first the multiplicative inverse, then the affine transformation. Also traditionally they are using the 16×16 ready S-Box.

ShiftRow(): this function takes the output of the subByte function and applies normal row shifting on the three last rows of the state's table.

MixColumn(): the input of this function is the output of the ShiftRow function. It uses matrix multiplication. And result is four bytes as byte for each row.

Table I summarize the device utilization of DYN SBOX 5STAGE for the target device XC7z030-2fbg676. The most consumed resource is the Number of fully used LUT-FF pairs. The utilization summary in table I, is only for the DYN SBOX 5STAGE, it has 18 bonded IOBs as shown in the Top Level Module of DYN SBOX 5Stage given in Fig. 5. the 18 IOBs are 8-bits of BYTE_IN, RST, and SYS_CLK as inputs, and 8-bits of SUB_BYTE as output. The main function of this unit is byte substitution.

TABLE I. DEVICE UTILIZATION SUMMARY FOR DYN SBOX 5STAGE

Device Utilization Summary			
Slice Logic Utilization	Used	Available	Utilization
Number of Slice Registers	55	157,200	1%
Number of Slice LUTs	56	78,600	1%
Number used as logic	56	78,600	1%
Number of occupied Slices	31	19,650	1%
Number with an unused Flip Flop	23	71	32%
Number with an unused LUT	15	71	21%
Number of fully used LUT-FF pairs	33	71	46%
Number of slice register sites lost to control set restrictions	1	157,200	1%
Number of bonded IOBs	18	250	7%
Number of BUFG/BUFGCTRLs	1	32	3%
Average Fanout of Non-Clock Nets	3.29		

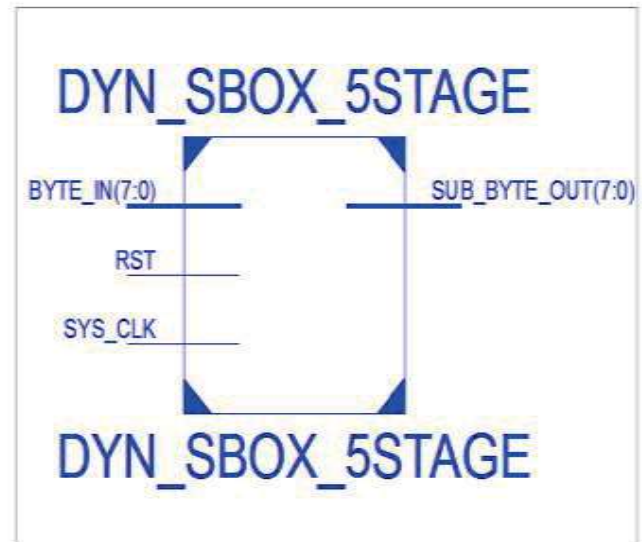


Fig. 5. Top Level Module of DYN SBOX 5Stage

The RTL schematic of the DYN SBOX 5STAGE unit is shown in Fig. 6. most of the blocks in this Fig. are LUT-FF pairs. The target device is from Xilinx Zynq-7000 All Programmable SoC family as shown in Fig. 7.

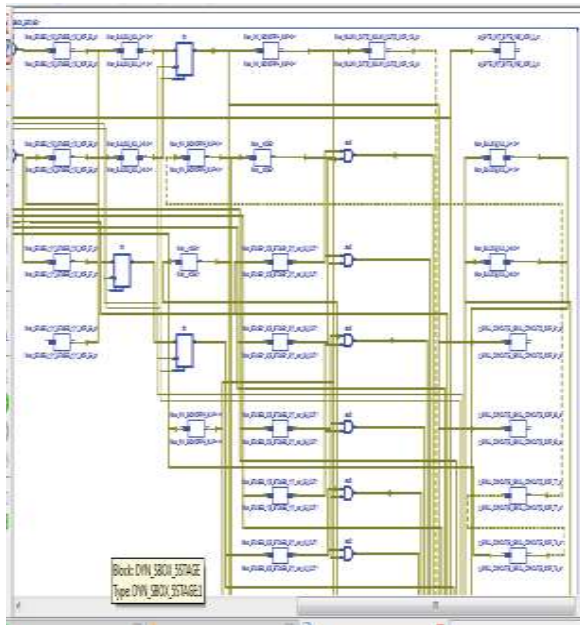


Fig. 6. RTL Schematic of of DYN SBOX 5Stage

Fig. 8. shows the simulation result of the AES Byte Substitution Transformation. As shown from the S-box table in [4], if the input from state is "00" the output is "63", where it uses the intersection of the most significant 4 bits as the row value and the Low significant 4-bits as the column value. Every byte needs 1 clock cycle. So for full state of 16 values it needs 16 clock cycles.

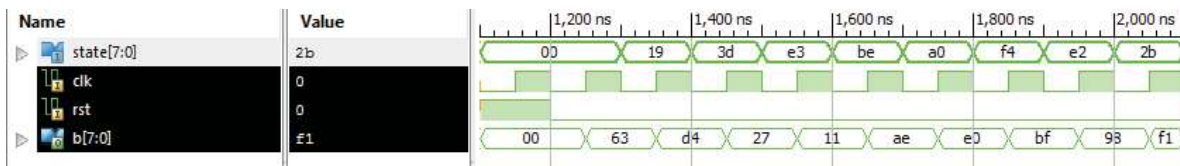


Fig. 8. Simulation Results of Byte Substitution Transformation

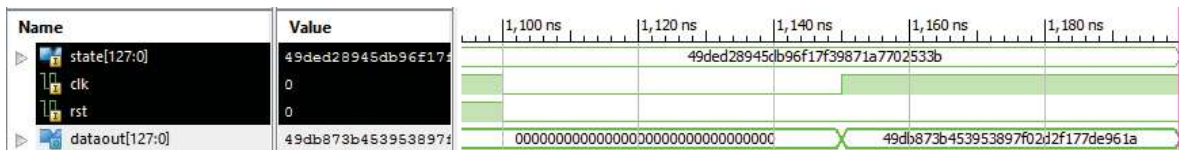


Fig. 9. Simulation Results of Shift Row Transformation



Fig. 7. Xilinx Target Device and used tools

Fig. 9. shows the simulation of ShiftRow transformation. Having the state value 49ded28945db96f17f39871a7702533b the output shows that the first row remains the same and the shifting process starts from the second row as shown in the output result. The state reads column by column so the first row value is 49 45 7f 77, it remains the same in the output.

Fig. 10. shows the simulation of Mix column transformation that is done by implementing the following equations as in [4]:

$$\begin{aligned} S'_{0,e} &= ((02) \cdot S_{0,e}) \oplus ((03) \cdot S_{1,e}) \oplus S_{2,e} \oplus S_{3,e} \\ S'_{1,e} &= S_{0,e} \oplus ((02) \cdot S_{1,e}) \oplus ((03) \cdot S_{2,e}) \oplus S_{3,e} \\ S'_{2,e} &= S_{0,e} \oplus S_{1,e} \oplus ((02) \cdot S_{2,e}) \oplus ((03) \cdot S_{3,e}) \\ S'_{3,e} &= ((03) \cdot S_{0,e}) \oplus S_{1,e} \oplus S_{2,e} \oplus ((02) \cdot S_{3,e}) \end{aligned}$$

After each multiplication the result is checked. if the MSB in the result is '1' then the result is XORed with "1b" in hex otherwise no XORing is needed.

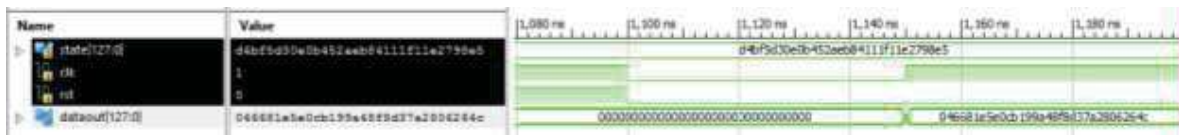


Fig. 10. Simulation Results of MixColumn Function

VI. CONCLUSION

This paper reviewed different types of security threats in healthcare industries. And it reviewed some of the security countermeasures for these types of threats. From literature, Encryption is considered to be the most important countermeasure. Also this paper reviewed some of the most common encryption methods that are used as basis for producing new methods that suits the dynamic IoT framework. These algorithms are DES, 3DES, bluefish and AES. And according to different studies reviewed AES algorithm is the most highly secured and the fastest encryption algorithm. This is more suitable to be used in the critical types of applications like healthcare application. Moreover this paper shows the implementation and simulation of the costliest component in AES algorithm (s-box). It is implemented using the ready s-box a LUTs. As a future work this component will be optimized in term of it design complexity.

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